PLAY BY PLAY: ADMINISTRATIVE INSIGHTS

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COMPUTER SCIENCE

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Purpose

- A web-based admin dashboard designed to manage a real-time sports prediction game
- Allows administrators to control live games while players interact on mobile devices
- Supports creating games, asking questions, and managing outcomes in real time
- Helps track player participation, engagement, and overall game performance

Tech Stack

- React & TypeScript (Frontend)
- Firebase / Firestore (Backend)
- Firebase Authentication (Admin Access)
- Vitest (Testing – 800+ tests)
- Vite (Build Tool)
- CSS with theme support

Core Featues

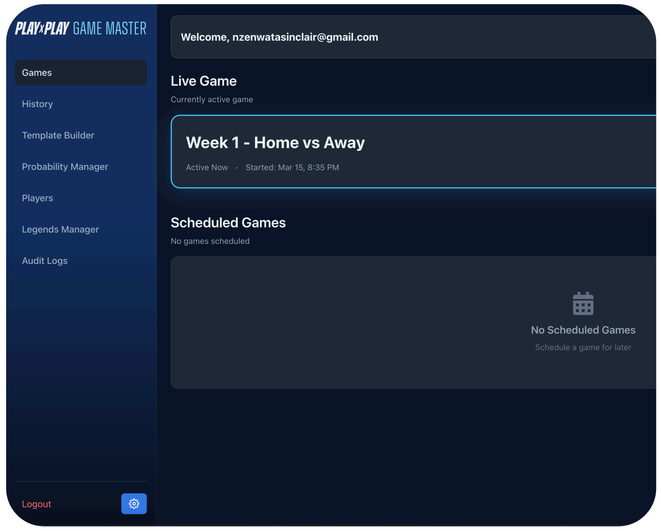
1. Game Management Dashboard
   Create and manage live games, open/close question windows, and select correct answers in real time
2. Question & Prediction Control
   Add and manage questions during gameplay, control response windows, and finalize results
3. Legends Mode (Historical Games)
   Convert past games into "legend" experiences so users can replay and interact with previous matches
4. Game History & Insights
   Search past games and review player participation, questions asked, and overall activity
5. Player Search & Management
   Quickly find players by username, email, or ID to view and manage their data

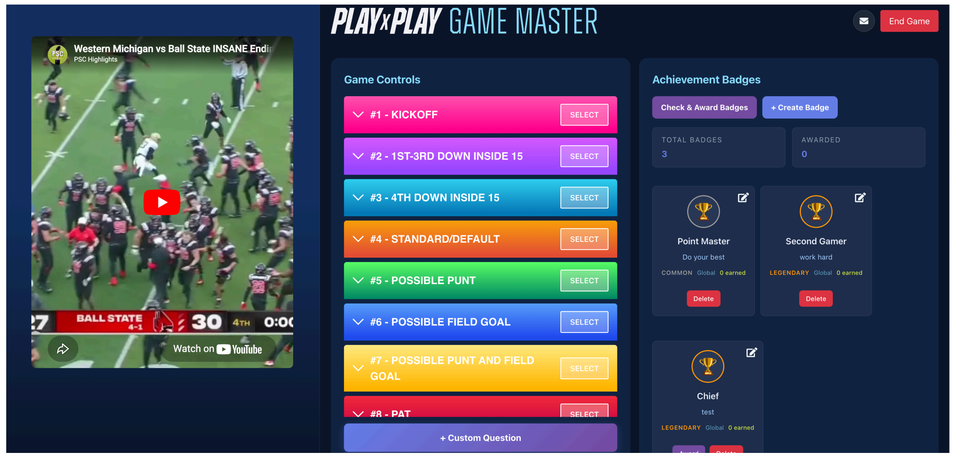
Additional Features

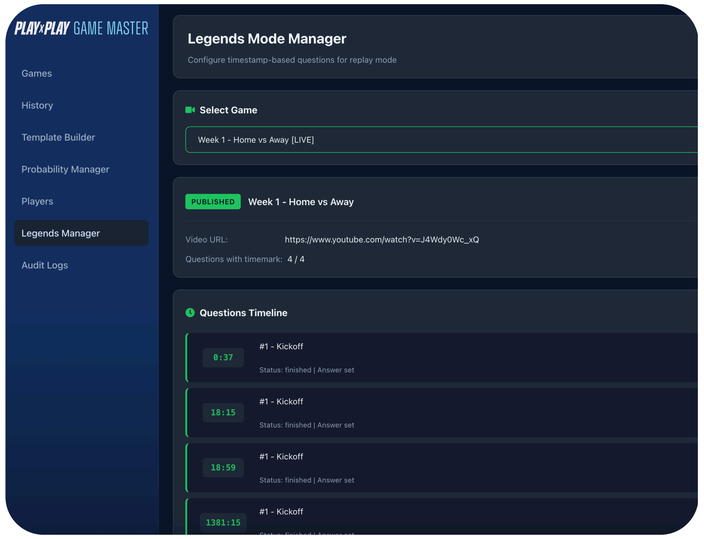
- Template Management
  Create, edit, and reuse game templates with predefined questions to speed up game setup
- Theme & Accessibility Support
  Switch between light, dark, and high-contrast modes for improved usability
- Audit Logging
  Track admin actions with timestamps to maintain transparency and accountability
- Live Announcements
  Send real-time messages and updates to users during gameplay

System Design

- Built with a modular structure for easy updates and scalability
- Separates frontend and backend logic for better maintainability
- Optimized data handling to ensure smooth real-time performance
- Designed to support high user interaction during live games







Conclusion

This project demonstrates how a real-time administrative system can effectively manage interactive sports prediction games. By combining live game control, historical replay features, and user management tools, the platform improves both administrative efficiency and player engagement. The system highlights the importance of usability, scalability, and reliability in building real-world applications.